

AI Skin Intelligence & Personalized Skincare Planner

AI-Powered Personalized Skin Assessment, Routine Planning, Ingredient Intelligence and Product Recommendation Platform

Project Overview: Skincare selection varies between individuals because human skin responds dynamically to **skin type** (oily, dry, combination, sensitive), active **skin concerns** (acne, hyperpigmentation, redness), individual **sensitivity thresholds**, daily **lifestyle stress**, **sleep quality**, fluid **hydration**, **environmental exposure** (UV index, ambient climate), and personal **preferences**. Generic over-the-counter advice fails to unify these heterogeneous factors. The AI Skin Intelligence platform bridges this divide by combining computer vision skin assessment with personalized profiling, active ingredient intelligence, and chronobiological routine planning.

Problem Statement: Generic skincare advice ignores individual biology, triggering product abandonment, wasted spending, or acute skin barrier damage.	Proposed Solution: Combines EfficientNet-B0 computer vision with lifestyle profiling, ingredient safety rules, and budget-tailored product recommendations.
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Project Objectives

- **Analyze Skin Images:** Classify skin conditions via EfficientNet-B0 PyTorch CNN.
 - **Build Personalized Profile:** Synthesize skin type, concerns, sleep, and hydration.
 - **Generate Skincare Routines:** Build dynamic morning, evening, weekly, and seasonal tiers.
- **Check Ingredient Compatibility:** Flag biochemical conflicts (e.g. Retinol + BHA).
 - **Recommend Matched Products:** Filter catalog by suitability score, budget, and allergies.
 - **Track Progress & Reports:** Calculate 5-factor health score; stream PDF/CSV/XLSX exports.

Platform Feature Matrix

Authentication & Security JWT HS256, Google OAuth, verified tokens.	AI Skin Assessment EfficientNet-B0 across 8 clinical categories.	User Skin Profile Dermatological baseline, habits, climate.
Personalized Routines 5 Tiers: AM, PM, Weekly, Monthly, Seasonal.	Ingredient Intelligence Biochemical safety & conflict detection.	Product Recommendations Catalog matching with allergy penalties.
Skin Health Score 5-Factor weighted formula (0–100).	Analytics & Diary Adherence trends, timeline, photo tracking.	Clinical Reports & Alerts PDF/CSV/XLSX streams, reminder alerts.
Role-Based Access (RBAC) 4 Roles: USER, CONSULTANT, DERM, ADMIN.	Admin Telemetry & Logs Audit logging, user controls, live telemetry.	Docker Containerization 3-container stack: NGINX, FastAPI, PostgreSQL.

Implemented Production Technology Stack

Layer	Technology	Implementation Details & Production Role
Frontend UI	React 18 + Vite	Modular Single Page Application, responsive CSS design tokens, Axios client.
Backend API	FastAPI (Python 3.11)	Asynchronous REST API, Pydantic v2 schemas, automated OpenAPI/Swagger docs.
AI / ML Engine	PyTorch + EfficientNet-B0	Transfer-learned CNN, CPU inference (~29ms latency), 8 skin concern classes.
Database & ORM	PostgreSQL 15 + SQLAlchemy	18 relational tables, connection pooling, ACID compliance, Alembic migrations.
Authentication	JWT + Google OAuth	HS256 access tokens, cryptographic OAuth2 ID token verification, RBAC guards.
Web Server & Gateway	NGINX Alpine	Reverse proxy on port 80, static SPA bundle serving, security header injection.
Containerization	Docker & Docker Compose	Multi-stage lean containers, unprivileged appuser (UID 10001), healthchecks.

System Architecture Dataflow

[1] USER	[2] REACT FRONTEND	[3] NGINX GATEWAY	[4] FASTAPI BACKEND	[5] AI + RULES ENGINE	[6] POSTGRESQL DB
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Academic Project Context & Evaluator Overview

- **Academic Presentation:** Capstone Engineering Project prepared for College Mentor review.
- **Engineering Principles:** Defensive architectural design, zero fabricated claims, strict data isolation, and Docker containerization.
- **Verification Level:** 100% verified local Docker multi-container stack, 161 backend pytest tests passing, and 36 unified phase checks passing.

User Journey: Login, Verification and Profile

Comprehensive onboarding workflow establishing authenticated identity, verified channels, and personalized baseline parameters.

Application Entry & Authentication Flow

[1] User Browser	[2] Open Application	[3] Login / Register	[4] Authentication (JWT)	[5] User Dashboard
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Registration, Login & Google OAuth Federation

User Registration: Standard self-registration requires full name, email, and password. The backend strictly defaults to role='USER' (privileged roles like CONSULTANT or ADMIN cannot be self-assigned). Passwords are cryptographically hashed via PBKDF2-HMAC-SHA256 (100,000 iterations, 16-byte random salt). Accounts initialize with email_verified=False.

Credential Login: Validates email and timing-safe password hash. Issues an HS256 JWT access token (1440-min expiry) and refresh token. Protected endpoints authenticate via Authorization: Bearer <token>. Suspended users are immediately blocked via is_blocked.

Google OAuth Federation: Verifies Google ID tokens cryptographically, extracting verified email and subject ID. (Note: Google OAuth in production requires external Google Client ID and domain configuration; dev mock fallback active for tests.)

Multi-Channel Identity Verification

Channel	Security Architecture & Implementation Details	Operational Boundary & Status
Email Verification	Generates a single-use 32-byte cryptographic token (SHA-256 stored in email_verification_tokens). 24h expiry prevents replay attacks. Rate-limited to 1 request per 60s to block mailbox flooding.	Active / Verified Token lifecycle verified.
Phone (SMS) OTP	Issues a secure 6-digit numeric OTP (SHA-256 stored in phone_otp_verifications). 10-minute expiration, max 5 incorrect attempts, max 3 requests per 10 minutes.	Configuration Required Real SMS delivery requires external Twilio configuration.

User Profile Model & Personalization Inputs

Downstream personalization relies on 4 baseline profile dimensions captured during onboarding in skin_profiles:

Dermatological Baseline - Skin Type: Dry, Oily, Combination, Normal, Sensitive - Concerns: Acne, Redness, Hyperpigmentation, Dryness - Allergies & Sensitivities (Fragrance, Essential Oils)		Lifestyle & Physiological Habits - Hydration: Daily fluid volume in liters (target: 2.0L+) - Sleep Quality: Nocturnal rest supporting cell repair - Stress Level: Cortisol impact on skin lipid barrier	
Environmental Exposure - Solar UV Index: Daily outdoor sun exposure hours - Local Climate: Arid, Humid, Temperate, or Cold - Urban Pollution: Environmental oxidative stress		Regimen Preferences & Constraints - Budget Tier: Low (<INR 1500), Medium, or Premium - Texture Preferences: Gel, Cream, Serum, Balm - Active Tolerance: Frequency limits for retinoids/acids	
[1] USER PROFILE	[2] LIFESTYLE HABITS	[3] ENVIRONMENTAL CONTEXT	[4] PERSONALIZATION INPUTS

Centralized User Dashboard Features

Implemented Dashboard Capabilities Real-Time Skin Health Score: Visual gauge displaying overall score and individual 5-factor component breakdowns. Active Routine Quick-Access: Direct view of Morning (cleanser, serum, SPF) and Evening (cleanse, repair, balm) protocols. Diagnostic Actions: Instant buttons to launch AI Skin Assessment, verify ingredient compatibility, and browse matched products. Compliance Diary & Alerts: Daily routine checkbox logging, hydration goal tracking, and notification center inbox.
Privacy, Data Isolation & Boundary Enforcement Strict Tenant Isolation: Database foreign keys bind all assessments, routines, and logs to authenticated user_id. Cross-User Protection: User A cannot access or modify User B's diagnostic records. Confirmed by integration tests. Protected Telemetry: Telemetry endpoints strictly reject anonymous (401) and standard user (403) calls; only ADMIN permitted.

AI Skin Assessment & Skin Intelligence

Deep learning computer vision evaluates dermatological skin condition categories to augment personalized planning.

Computer Vision Assessment Pipeline

Image Ingestion: The user captures or uploads a close-up facial photograph via web file picker or webcam (JPEG, PNG). The image is submitted to the inference endpoint (/api/image-analysis/upload or /api/assessments/scan):

[1] SKIN IMAGE	[2] PREPROCESSING	[3] EFFICIENTNET-B0	[4] PYTORCH INFERENCE	[5] 8 CATEGORIES	[6] ASSESSMENT RESULT	[7] PERSONALIZATION ENGINE
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Model Architecture & Configured Classes

Specification	Technical Detail	Dermatological Category Classes (8 Configured Classes)
Architecture	EfficientNet-B0 (Inverted Bottleneck MBConv)	1. Acneiform & Follicular Disorders (Acne, Folliculitis)
Framework	PyTorch 2.x (Torchvision, CPU-optimized)	2. Eczematous & Inflammatory Dermatitis (Eczema, Atopic)
Model Checkpoint	skin_condition_improved.pth (15.61 MB)	3. Infections & Infestations (Fungal, Bacterial, Viral)
Model Metadata	improved_model_metadata.json (v2.0.0)	4. Other Clinical Disorders
Input Dimensions	224 × 224 × 3 (Normalized to ImageNet mean/std)	5. Papulosquamous Disorders (Psoriasis, Lichen Planus)
Model Footprint	5.3M Parameters (Lean, efficient CPU execution)	6. Trauma & Insect Bites
Execution Mode	CPU-supported inference (device-agnostic loader)	7. Urticaria & Reactive Rashes 8. Vascular & Purpuric Conditions (Rosacea, Erythema)

Synthesis with User Profile & Lifestyle Context

Computer vision classifications are synthesized with questionnaire responses to eliminate optical artifacts (e.g. poor lighting, camera glare):

Multimodal Synthesis Formula:

Personalized Skincare Intelligence = AI Assessment + User Profile + Lifestyle Rhythms + Sleep + Hydration + Environment

If the model detects surface oiliness but the profile reveals chronic water deficiency and dry air conditioning, the engine identifies surface dehydration overproduction and prescribes hydrating humectants (Hyaluronic Acid) rather than harsh drying agents.

Empirical Model Metrics & Docker Benchmark

Performance Dimension	Verified Value	Source / Environment	Scientific Classification / Label
Validation Accuracy	84.2%	improved_model_metadata.json	Reported Model Validation Metric
Macro F1-Score	0.825	improved_model_metadata.json	Reported Model Validation Metric
CPU Latency (P50)	29.45 ms	Docker Backend Container (30 runs)	Local Docker Benchmark
CPU Latency (P95)	36.07 ms	Docker Backend Container (30 runs)	Local Docker Benchmark
Inference Determinism	100.0%	Identical outputs on fixed tensor input	Engineering Verification

Note: Reported metrics reflect model validation on the SCIN dataset. Independent clinical/test-dataset validation is not established. Local Docker benchmark reflects containerized CPU execution.

MANDATORY CLINICAL SAFETY NOTICE & SCOPE LIMITATION

The AI assessment is intended to support personalized skincare planning and does not replace professional medical evaluation. The platform provides AI-assisted cosmetic skin-condition assessment and decision-support guidance. It does NOT provide clinical dermatological diagnosis, oncology screening, or medical prescriptions. Users with persistent rashes, atypical lesions, or severe inflammatory conditions must consult a board-certified dermatologist.

Personalized Skincare Intelligence

Translating multidimensional assessment data into chronobiological routines, ingredient safety rules, and curated products.

Personalization Engine Architecture

[1] USER PROFILE	[2] AI ASSESSMENT	[3] LIFESTYLE	[4] SKIN CONCERNS	[5] RULES ENGINE	[6] ROUTINES & PRODUCTS
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Multi-Tier Chronobiological Skincare Routines

Routine Tier	Chronobiological Objective	Step Sequence & Formulation Order
Morning (AM)	Antioxidant defense, moisture retention, broad-spectrum UV shielding.	1. Cleanse: Gentle Cleanser → 2. Treatment: Niacinamide 5% + HA → 3. Moisturize: Ceramide Cream → 4. Protect: Mineral SPF 50+
Evening (PM)	Deep cleansing, targeted cellular turnover, barrier restoration.	1. Double Cleanse: Micellar Oil + Gentle Cleanser → 2. Targeted Active: Retinol 0.3% OR Azelaic 10% → 3. Repair: Lipid Night Balm
Weekly	Stratum corneum exfoliation & intensive moisture replenishment.	1–2× weekly: Gentle Lactic/Salicylic Acid exfoliation OR Deep Hydration Mask (scheduled strictly on non-retinoid nights).
Monthly	Periodic barrier audit, tolerance check, and active progression.	Skin diagnostic re-assessment questionnaire, pore purifying clay treatment, and active strength progression review.
Seasonal	Adapting to temperature, ambient humidity, and UV index shifts.	Winter: Shift to richer lipid occlusives; Summer: Shift to non-comedogenic lightweight gels and sweat-resistant SPF 50+.

Adaptive Regimens & Manual Customization

Routines adapt dynamically upon re-assessment: **(a) Acne Delta:** If acne improves, active treatment frequency is reduced from 3x to 2x/week; if increased, targeted treatment is prioritized; **(b) Sensitivity Delta:** If redness spikes, exfoliating acids are paused in favor of barrier repair lipids (Ceramides, Panthenol); **(c) Manual Editing:** Users can customize step details, frequencies, and instructions via PUT /api/routines/{id}.

Ingredient Intelligence & Contraindication Matrix

Active Ingredient Pair	Safety Status	Biochemical Interaction & Implemented Safety Rule
Retinol + Salicylic Acid (BHA)	INCOMPATIBLE / UNSAFE	Implemented Rule: Direct concurrent layering causes acute barrier exfoliation, peeling, and irritation. The engine flags this pair as unsafe and splits them into alternating evenings.
Niacinamide + Hyaluronic Acid	SAFE & SYNERGISTIC	Implemented Rule: Highly synergistic. Hyaluronic acid supplies humectant hydration while Niacinamide upregulates ceramide synthesis and calms redness.
Vitamin C + AHA/BHA	CAUTION / SEPARATION	Implemented Rule: Both require low acidic pH; concurrent application triggers stinging. Recommended for separation: Vitamin C in AM, AHA/BHA in PM.

Note: Implemented rule engine covers primary dermatological contraindications. Does not claim exhaustive coverage of all chemical compounds.

Product Recommendation Engine vs Product Recognition

[1] USER PROFILE	[2] SKIN CONCERNS	[3] ROUTINE STEPS	[4] INGREDIENT RULES	[5] BUDGET CEILING	[6] PRODUCT CATALOG	[7] RECOMMENDED PRODUCTS
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CRITICAL ARCHITECTURAL DISTINCTION: RECOMMENDATION VS RECOGNITION

- Product Recommendation (IMPLEMENTED & VERIFIED):** The recommendation engine evaluates user profile needs, active skin concerns, and budget tiers against a verified catalog of skincare formulations. Formulations are ranked on a 0–100 suitability scale (Skin Type: 35pts, Concerns: 35pts, Rating: 15pts, Synergy: 15pts), with strict **-40 point penalties** for user allergens.
- Product Recognition (OUT OF CURRENT SCOPE):** Identifying physical commercial product bottles from optical camera photographs is a separate computer vision task and is **NOT** implemented in this project.

Skin Health Score, Progress Tracking & Reports

Empirical 5-factor wellness scoring, longitudinal diary analytics, and multi-format clinical report generation.

Mathematical Formulation of the Skin Health Score

The platform computes a deterministic **Skin Health Score** bounded strictly between 0 and 100. The score is derived from a 5-factor weighted model verified in `app/services/skin_health_scoring.py`:

5-Factor Skin Health Score Equation			
Skin Health Score = [0.35 * Condition] + [0.20 * Lifestyle] + [0.15 * Sleep] + [0.20 * Routine] + [0.10 * Hydration]			
Score Dimension	Weight	Evaluation Metric & Derivation	Personalization & Baseline
Skin Condition	35% (0.35)	AI concern severity, acute erythema, dryness, and barrier reactivity.	Dry skin prioritizes moisture dehydration; Sensitive skin prioritizes erythema/reactivity.
Lifestyle Rhythms	20% (0.20)	Stress level, physical activity frequency, and environmental pollution.	Elevated stress penalizes recovery; active lifestyle boosts microcirculation.
Sleep Quality	15% (0.15)	Nocturnal sleep duration against 7–9 hour cellular regeneration window.	Penalizes chronic sleep deficit (<6h) which impairs epidermal barrier repair.
Routine Consistency	20% (0.20)	14-day compliance adherence: Morning SPF and evening cleanse/treatment.	Derived from <code>skincare_logs</code> : $(\text{completed_logs} / \text{total_logs}) * 100$.
Hydration Level	10% (0.10)	Daily fluid intake evaluated against 2.0–2.5 L/day dermatological guideline.	Linear scale: $\text{water_intake} / 2.0 * 100$, clamped to 100.0 max.
Total System Weight	100% (1.00)	Bounded strictly to [0.0, 100.0]. Missing values use neutral baseline handling.	Intended for wellness tracking, NOT a clinical diagnosis.

IMPORTANT: Termed 'Skin Health Score', NOT 'Clinical Health Score'. Intended for personalized wellness and tracking.

Longitudinal Analytics & Progress Diary

Epidermal barrier restoration follows a 28-to-40-day biological cycle. The analytics engine (`/api/analytics/*`) tracks longitudinal improvements across three verified modules:

[1] SKIN ASSESSMENT	[2] STORED HISTORY (DB)	[3] ANALYTICS ENGINE	[4] PROGRESS TIMELINE	[5] USER INSIGHTS
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- **Historical Assessment Trends:** Visual tracking of overall score trajectory and individual concern deltas (acne, redness, dryness).
- **Routine Compliance Logs:** Daily morning and evening checklist logging in `skincare_logs` measuring regimen adherence rate.
- **Progress Photo Diary:** User diagnostic photographs stored in `skin_progress_photos` linked to assessments for visual comparison.

Multi-Format Clinical Reports & Notification Engine

Export Format	MIME Type / Engine	Document Contents & Intended Clinical Use Case	Verification Status
PDF Clinical Report	application/pdf ReportLab 5.x	Structured clinical summary: profile overview, AI concern breakdown, morning/evening routine tables, and active ingredient list.	Verified Stream P50 = 49.8ms in Docker
CSV Timeseries	text/csv Streaming response	Tabular dump of historical skin scores, hydration entries, sleep metrics, and routine compliance dates for spreadsheet analysis.	Verified Stream 13.1 KB payload verified
XLSX Workbook	application/vnd.ms-excel OpenPyXL	Multi-tab structured workbook separating user profile, historical assessments, routine steps, and recommended product catalog.	Verified Stream 19.0 KB payload verified

Notification Center: In-app alerts notify users of Morning/Evening routines, hydration targets, and seasonal profile reviews. External SMS/email delivery requires configured third-party providers (Twilio / SendGrid); operates in console fallback when unconfigured.

Security, Roles & Technical Architecture

Hardened multi-tier architecture implementing defense-in-depth, strict RBAC, and relational database modeling.

Role-Based Access Control (RBAC) Architecture

Role Name	Authorized Capabilities & Purpose	Security Enforcement & Scope
USER	Personal skin assessment, routine generation, product browsing, personal analytics, and PDF reports.	Strictly isolated to owned records via user_id database foreign keys.
SKINCARE_CONSULTANT	Authorized review of assigned client skin profiles, custom routine overrides, and consultation notes.	Requires role claim; access restricted to explicit client-consultant assignments.
DERMATOLOGIST	Authorized clinical workspace access, review of AI diagnostic categories, and clinical reviews.	Requires verified practitioner status; manages clinical review workflows.
ADMIN	Administrative system operations: user account controls, telemetry inspection, and audit logging.	Full platform governance; protected by require_roles('ADMIN') middleware.

Security Architecture & Multi-User Isolation

[1] USER	[2] AUTHENTICATION	[3] JWT VALIDATION	[4] ROLE CHECK (RBAC)	[5] PROTECTED API	[6] USER-OWNED DATA
JWT Authentication & Password Security - JWT tokens signed with HS256 algorithm and 1440-minute expiry. - PBKDF2-HMAC-SHA256 password hashing with 100k rounds & salt. - Google OAuth2 ID token cryptographic backend signature verification.			Telemetry Hardening & Endpoint Protection /api/system/telemetry strictly restricted to authenticated ADMIN. - Rejects anonymous (401) and standard user (403) requests. - Rate limiting on contact OTP endpoints prevents brute-force attacks.		
Multi-User Isolation (Tenant Boundaries) - User A (user_id=1) can never read or mutate User B's records. - Foreign key cascades isolate assessments, routines, logs, and photos. - Cross-tenant penetration verified: returns 404/403 on illegal access.			Container & HTTP Infrastructure Defense - Backend runs under unprivileged appuser (UID 10001). - NGINX headers: X-Frame-Options: DENY, nosniff, XSS-Protection: 1. - Read-only bind mount for ML model weights prevents runtime tampering.		

PostgreSQL Relational Database Schema (18 Tables)

Cluster Domain	Actual Database Tables (from models.py)	Relational Integrity & Foreign Keys
Identity & Auth	users, email_verification_tokens, phone_otp_verifications, admin_audit_logs	Unique email index, password hashes, foreign key cascade to User.
Profile & State	skin_profiles	1-to-1 relationship with User; stores baseline skin, habits, and climate.
AI & Assessment	skin_assessments, image_analyses	Stores raw class probabilities, concern tags, and calculated health score.
Routines & Plans	skincare_routines	1-to-many routine structure (AM, PM, Weekly, Monthly, Seasonal).
Ingredients	ingredients, ingredient_compatibility_checks	Active ingredient benefits, side effects, conflicts, and audit log checks.
Product Catalog	products, product_recommendations	Catalog items, suitability scores, purchase links, and budget tiers.
Diary & Analytics	skincare_logs, skin_progress_photos	Daily morning/evening checklist adherence and historical progress photos.
Clinical Workspace	consultations, clinical_reviews	Patient-consultant bookings, clinical notes, and routine overrides.
Notifications	notifications, reminder_settings	Categorized alerts (ROUTINE, HYDRATION, SLEEP) and daily reminder times.

Physical Tiered System Architecture

[1] USER BROWSER	[2] REACT + VITE SPA	[3] NGINX (PORT 80)	[4] FASTAPI (PORT 8000)	[5] AI MODEL / RULES	[6] POSTGRESQL (PORT 5432)
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External Integrations (where configured): Google OAuth2 Federation, Email SMTP Gateway, Twilio SMS Provider, Cloud Object Storage.

Complete Project Workflow, Docker & Verification

Full 18-step execution lifecycle, verified multi-container Docker topology, and empirical verification scorecard.

Complete 18-Step End-to-End Execution Workflow

1. Open Application User loads web client; NGINX serves Vite React SPA on port 80.	2. Register / Login User authenticates via email/password or OAuth; receives JWT token.	3. Verify Identity Email verification token and phone OTP safeguard user accounts.
4. Create Skin Profile User logs baseline skin type, tone, concerns, and sensitivities.	5. Enter Lifestyle Info User enters daily water intake, sleep quality, stress level, and UV.	6. Upload Skin Image User captures or uploads a close-up facial photo for AI assessment.
7. AI Skin Assessment EfficientNet-B0 infers probabilities across 8 condition categories.	8. Generate Concerns Personalization engine joins AI findings with questionnaire targets.	9. Calculate Health Score Deterministic 5-factor formula computes [0–100] barrier score.
10. Generate Routines Engine builds morning, evening, weekly, monthly, seasonal tiers.	11. Check Compatibility Rule engine checks active pairs, flagging Retinol + BHA conflicts.	12. Recommend Products Catalog matching ranks items by suitability, budget, and allergies.
13. Track Daily Progress User logs routine step execution and progress photos in daily diary.	14. View Analytics User tracks longitudinal score trends, acne delta, and adherence.	15. Generate Reports User or clinician exports clinical summaries via PDF, CSV, or XLSX.
16. Receive Notifications In-app alerts remind users of morning SPF, water, and evening care.	17. Adapt Routines Re-assessments recalibrate active treatment frequencies dynamically.	18. Long-Term Care User maintains barrier health with routine consistency and reviews.

Docker Architecture & Verification Scorecard

[1] DOCKER COMPOSE		[2] PostgreSQL 15 Container		[3] FastAPI Backend Container		[4] React + NGINX Container		[5] PyTorch AI Model Loading	
Subsystem / Component		Status		Empirical Verification Details & Operational Boundary					
Docker Compose Stack		VERIFIED		3 containers (db, backend, frontend) healthy; internal bridge network; healthchecks pass.					
FastAPI Backend & NGINX		VERIFIED		Python 3.11-slim, unprivileged appuser (UID 10001); NGINX reverse proxies port 80 to 8000.					
PostgreSQL Database		VERIFIED		PostgreSQL 15-alpine healthy; 18 tables initialized; persistent volume storage verified.					
AI Model Loading & Inference		VERIFIED		EfficientNet-B0 loaded on CPU; P50 = 29.45ms, P95 = 36.07ms latency; 8 classes classified.					
Authentication & RBAC		VERIFIED		JWT HS256 auth, 4 roles enforced, tenant data isolation verified (401/403/200 tests pass).					
Routines & Ingredients		VERIFIED		5 routine tiers; dynamic adaptation; contraindications flag Retinol + Salicylic Acid.					
Recommendations & Score		VERIFIED		Catalog suitability ranking (0–100); 5-factor formula (0.35C+0.20L+0.15S+0.20R+0.10H).					
Analytics & Clinical Reports		VERIFIED		Diary logs and trend history; streamed PDF, CSV, and XLSX report downloads verified.					
Restart Recovery		VERIFIED		docker compose down followed by up -d restored all containers without data loss.					
External Cellular SMS		CONFIG REQUIRED		Console fallback active; real cellular SMS requires external Twilio SID credentials.					
Cloud & Clinical Validation		NEXT STAGE		Docker verified locally; cloud deployment (AWS/GCP) and clinical trials are future scope.					
Current Project Status: Local Docker runtime verification has been completed. The application, database, AI inference pipeline, authentication, authorization, personalization workflow, reporting, and major security controls have been verified in the local containerized environment. Cloud deployment, production-domain configuration, and external service validation are subsequent stages.				Future Project Scope: • Cloud deployment on AWS ECS / GCP Cloud Run with production HTTPS domains. • Integration of production Twilio SMS and transactional email gateways. • Scalable cloud object storage migration (AWS S3 / Google Cloud Storage). • Retraining on larger diverse datasets and independent clinical trials.					

FINAL PROJECT SUMMARY & CONCLUSION

The AI Skin Intelligence & Personalized Skincare Planner combines computer vision, user profiling, personalization logic, skincare ingredient intelligence, product recommendation, and progress tracking into one integrated platform. The system transforms user-specific information and AI-assisted skin assessment into personalized skincare planning while applying authentication, authorization, and data-isolation controls. The current local Docker environment has been verified, with cloud deployment and external service configuration remaining as the next stage.